Medical Diagnosis and Report Generation for Endoscopic Procedures Using Vision-Language Models

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B.Tech CSE Semester 8 Final Evaluation

Agenda

1. Introduction
2. Methodology
3. Architecture
4. Implementation
5. Results
6. Future Work
7. Conclusion

Introduction

The Clinical Burden

- Each endoscopic procedure requires a **detailed written report** — findings, diagnosis, and treatment recommendations
- Manual report writing takes **3–5 minutes per procedure**: a major bottleneck in high-volume GI units performing 30–50 cases per day
- A unit performing 40 procedures/day spends $\sim$**3.3 hours daily** on documentation alone — time taken directly away from patient care
- Repetitive documentation contributes to **physician burnout** and diagnostic fatigue Shanafelt et al. [1], Rotenstein et al. [2], National Academies of Sciences, Engineering, and Medicine [3]
- Documentation load has grown substantially with electronic health record requirements, doubling in the past decade National Academies of Sciences, Engineering, and Medicine [3]

Bottom line: Time spent writing reports is time *not* spent on patient care.

The AI Gap

- General-purpose MLLMs (GPT-4V, LLaVA, Gemini) perform well on natural images but **fail on specialised endoscopy**:
  - Cannot reliably identify subtle mucosal features (erythema, granularity, friability)
  - Unfamiliar with endoscopic grading systems (Paris classification, Barrett's Prague criteria, ESGE grades)
  - Produce generic descriptions rather than **structured clinical reports**
- Existing medical MLLMs (LLaVA-Med, HuatuoGPT-Vision) are trained on broad biomedical data — **not** on physician-annotated endoscopic reports
- **No model existed** trained on GMC Calicut physician-annotated GI endoscopic data, leaving a critical gap for Indian clinical settings

Our Objective

MVLMERG — Multimodal Vision-Language Model for Endoscopic Report Generation

Fine-tune a vision-language model to automatically generate professional-grade GI endoscopic reports.

- **Architecture:** LLaVA-4 base with CLIP-ViT-L/14 vision encoder + Vicuna-7B-v1.5 LLM backbone
- **Fine-tuning strategy:** P-LoRA — soft prompt tuning + selective LoRA
- **Two-stage training:**
  1. Stage 1: Medical domain knowledge injection (PubMed, MedTrinity-25M, VQA-Med, ∼60K GI records)
  2. Stage 2: Physician-style report generation (GMC Calicut, ∼31,200 annotated samples)
- **Target:** Automated, structured, interpretable endoscopic reports in ∼**7 seconds** vs. 3–5 minutes manual

Problem Definition

Core Problem:

- Writing endoscopic image reports requires deep domain expertise in gastroenterological terminology, lesion morphology, and clinical logic
- General Multimodal Large Language Models (MLLMs) exhibit critical knowledge gaps: they struggle with specialised endoscopy terminology, pathological feature descriptions, and complex medical reasoning
- Existing fine-tuning methods suffer from **limited domain adaptability** and **insufficient endoscopic-specific knowledge**
- High-quality physician-annotated endoscopic data is scarce — publicly available medical datasets (PubMed) cover general medicine but lack the specificity needed for endoscopic report generation

Goal: Develop **MVLMERG** — a fine-tuned vision-language model that bridges the domain gap, producing professional-grade endoscopic reports matching the standard of intermediate-level physicians. Ali et al. [4], Haghighi et al. [5]

Related Work: General Medical MLLMs

- **LLaVA-Med** Li et al. [6] — trained on PMC-15M biomedical image-text pairs; strong generalization across medical domains but **lacks endoscopy-specific terminology** and fine-grained lesion description
- **HuatuoGPT-Vision** — clinical knowledge via instruction tuning; good medical NLP but **insufficient for GI endoscopic findings**
- **BiomedGPT** — unified generative model across diverse biomedical tasks; **broad coverage but weak on GI endoscopic scenarios**
- **Uni-Med** — multi-task VLM covering radiology, pathology, dermatology; limited endoscopic performance

Common gap: Pre-training on broad medical corpora does not provide the specialised knowledge needed for *detailed* GI endoscopic report generation.

Related Work: Surgical / GI-Specific Models

- **SurgicalGPT** — fine-tuned GPT for surgical visual Q&A; 3–5% accuracy gain over generalist models but focuses on **surgical scenes**, not endoscopic mucosal analysis
- **ColonGPT / ColonINST** — colonoscopy instruction tuning with polyp detection focus; good for classification but **not designed for full report generation**
- **EndoChat** — endoscopy dialogue system; limited to conversational Q&A format, **cannot produce structured clinical reports**

What all GI-specific models lack:

- None trained on **physician-annotated, full-format** endoscopic reports
- None target **Indian clinical settings** (terminology, grading conventions differ)
- None support complete structured report generation (findings + diagnosis + recommendations)

Related Work: Fine-Tuning Methods

- **LoRA** Hu et al. [7] — Low-Rank Adaptation; adds rank-decomposition matrices $\Delta W = AB$ to frozen weights; highly parameter-efficient but **no mechanism to align input representations** with the target domain
- **Prefix / P-Tuning** Liu et al. [8] — soft prompt tokens prepended to input; aligns context but **without weight adaptation the model's knowledge remains unchanged**
- **PeFoMed** — parameter-efficient fine-tuning for medical LLMs; effective for general medical Q&A but **not evaluated on endoscopic scenarios**
- **MISS** — parameter-efficient medical MLLM fine-tuning; covers multiple imaging modalities but **not endoscopy-specific**

Key insight: Neither prompt tuning nor LoRA alone is sufficient — their **combination (P-LoRA)** addresses both input representation alignment *and* weight-space domain adaptation simultaneously.

Gap MVLMERG Fills

The Gap

No existing model combines (1) prompt-based input alignment, (2) selective weight adaptation, and (3) physician-annotated endoscopic training data.

- **P-LoRA** — unifies soft prompt tuning + selective LoRA: addresses both input alignment and domain knowledge injection in a single fine-tuning strategy
- **GMC Calicut physician-annotated data** — 6,500 real procedures annotated by senior endoscopists, augmented to ∼31,200 samples; reflects actual Indian clinical practice and terminology
- **Chain-of-Thought integration** — anatomical → visual → diagnostic reasoning baked into training data; produces interpretable, verifiable reports
- **First model** targeting professional Indian GI endoscopic report generation in a real hospital deployment setting

Methodology

Dataset Creation and Augmentation

- **Data Source:** Government Medical College (GMC), Calicut
- **Raw Data:** 6,500 **patient reports** (semi-structured PDFs + endoscopic images)
  - Each report includes **4 endoscopic images**
- **Data Augmentation:**
  - Applied **rotation and random translation** to every image
  - Purpose: increase dataset size and improve model robustness
  - **Approximate scale:**
    - 6,500 reports $\times$ 4 images $=$ 26,000 base images
    - Rotation alone multiplies by 4 $\rightarrow$ over 100,000 images
    - Translation variations further expand the pool
- **Text Data:**
  - Final image–report pairs after augmentation: $\sim$**31,200 samples**
  - Data split: 29,640 training / 1,560 validation
- **Multimodal Association:** Each Q&A pair linked to original + augmented endoscopic images Wang et al. [9], Zhang and et al. [10]

Figure 1: Custom Dataset Workflow

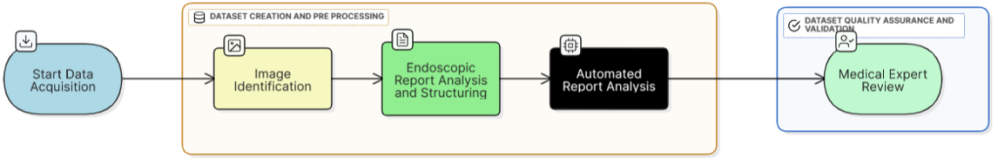


Figure: Data Workflow: From 6,500 patient reports to ∼31,200 image–report pairs Wang et al. [9], Zhang and et al. [10]

Data Augmentation: Rotation

- Each original endoscopic image is rotated by **0°, 90°, 180°, and 270°**
- Produces exactly **4 variants** per image
- Clinically safe: tissue textures and lesion appearances remain unchanged
- **Effect:** $4\times$ structured expansion of the base image set

Data Augmentation: Stochastic Translation

- **Sampling:** Bernoulli process with $p = 0.2$ selects approx. 20% of rotated images for translation
- **Shift model:** Translation vector (τ_x, τ_y) drawn from a truncated Gaussian:

$$\tau_x \sim \mathcal{N}(0, (0.05\,W)^2), \text{ clamped to } [-0.15\,W, +0.15\,W]$$
$$\tau_y \sim \mathcal{N}(0, (0.05\,H)^2), \text{ clamped to } [-0.15\,H, +0.15\,H]$$

- **Overlap guarantee:**

$$\text{Overlap Fraction} = \left(1 - \frac{|\tau_x|}{W}\right) \times \left(1 - \frac{|\tau_y|}{H}\right) \geq (1 - 0.15)^2 = 72.25\%$$

- **Result:** Controlled positional variation mimicking endoscopic camera jitter, without harming diagnostic content.

Data Augmentation Examples

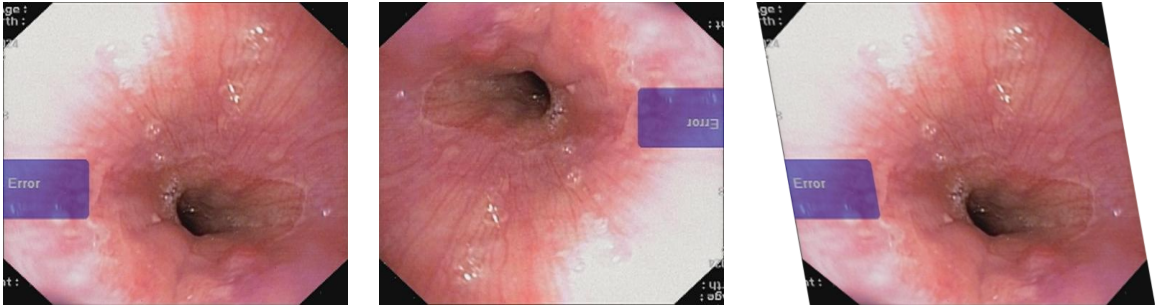
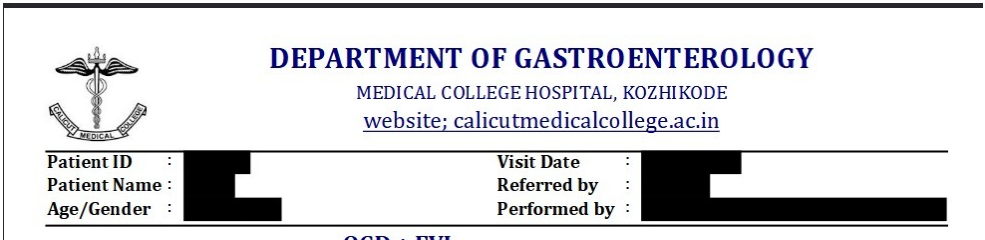


Figure: Left: Original Middle: Rotated (90°) Right: Randomly translated

Patient Data De-identification & Ethical Compliance

- **Regulatory Framework & Automated Script:**
  - Strict adherence to the **Digital Personal Data Protection (DPDP) Act**, supported by a formal MoU between GMC Calicut and NIT Calicut
  - **Redaction approach mirrors rigorous de-identification used in recent high-profile document releases (e.g., Epstein files)** – all personally identifiable fields irreversibly blacked out to ensure zero traceability
  - **Header Field Redaction:** Automatic blackout of Patient ID, Name, Age/Gender, Visit Date, UGI/OGD Numbers
- **Result:** Irreversible anonymization of all sensitive patient information before data enters the pipeline



DEPARTMENT OF GASTROENTEROLOGY

MEDICAL COLLEGE HOSPITAL, KOZHIKODE

website; calicutmedicalcollege.ac.in

Patient ID :	[redacted]	Visit Date :	[redacted]
Patient Name :	[redacted]	Referred by :	[redacted]
Age/Gender :	[redacted]	Performed by :	[redacted]

Report Analysis and Preprocessing

Medical Entities Extracted: Wang et al. [9], Zhang and et al. [10]

- Pathological findings and characteristics
- Diagnostic conclusions
- Procedural techniques and instruments

Processing Method:

- Natural language processing techniques
- Extraction of critical medical elements
- Foundation for Q&A pair generation

Automated Question-Answer Pair Generation

Multi-Stage Process: Pogorelov and et al. [11]

1. **Medical Entity Extraction** – Key concepts and diagnoses
2. **Question Template Design** – Diagnostic, anatomical, pathological, procedural, comparative
3. **Automated QA Generation** – Multiple question variations per concept Li et al. [6]
4. **Answer Synthesis** – Accurate answers from reports + medical knowledge

Result: Diverse, clinically relevant question-answer combinations.

Dataflow Pipeline and Methodology

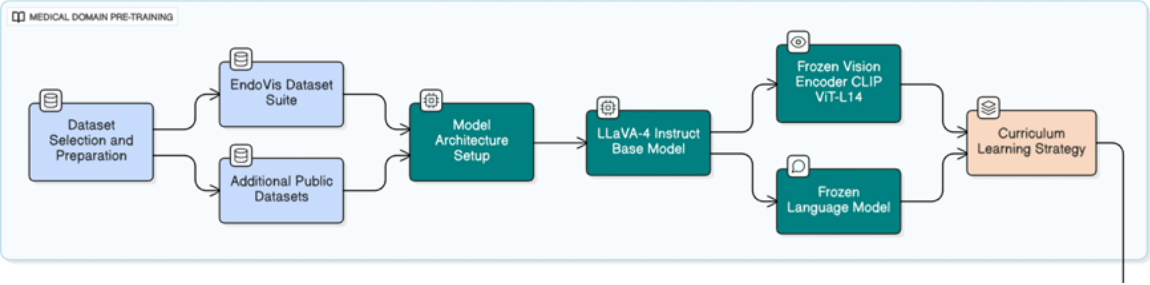


Figure: Stage 1: Pre-training.

Dataflow Pipeline and Methodology

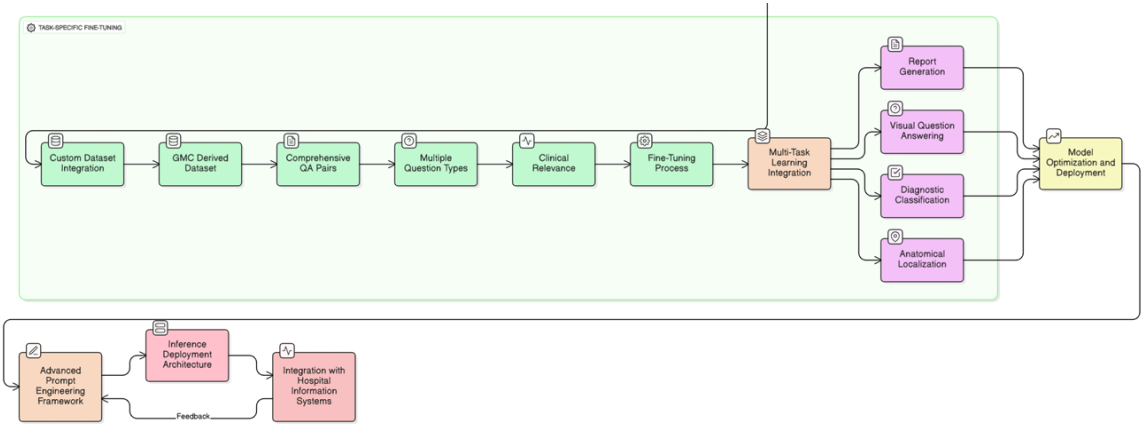


Figure: Stage 2: Fine-tuning

Architecture

Two-Stage Training Framework — MVLMERG

Stage 1: Medical Domain Knowledge Injection Leaman and Lu [12], Hu et al. [7]

- **Objective:** Acquire gastrointestinal domain knowledge
- **Base Model:** LLaVA-4 with frozen CLIP ViT-L/14
- **Trainable:** LLM backbone (vision encoder + connector frozen)
- **Datasets:** PubMed, MedTrinity-25M, VQA-Med ($\sim$60K GI records) Xie et al. [13], Ben Abacha et al. [14]

Stage 2: Professional Report Generation Leaman and Lu [12], Hu et al. [7]

- **Objective:** Learn physician-style endoscopic report language
- **Dataset:** GMC Calicut physician-annotated data (6,500 base; $\sim$31K after augmentation)
- **Strategy:** P-LoRA — prompt tokens + selective LoRA on key layers Hu et al. [7], Liu et al. [8]
- **Unfrozen:** LLM backbone (vision encoder + connector frozen)

P-LoRA: Prompt-based LoRA Fine-Tuning

Core Innovation of MVLMERG Hu et al. [7], Liu et al. [8]

P-LoRA combines **two complementary adaptation mechanisms**:

Soft Prompt Tuning

Prepends $m = 64$ learnable tokens to the input embedding sequence — steers the model's *attention* toward endoscopic domain context before any weight computation.

Selective LoRA

Injects low-rank adapter matrices $\Delta W = AB$ into the *first, middle, and last* transformer layers — adapts the model's *knowledge* to endoscopic terminology and clinical report style.

- Prompt tokens correct **input representation**; LoRA adapters correct **weight-space knowledge**
- Only **0.343%** of all model parameters are trainable
- **+5% CIDEr**, **+4.4% ROUGE-L** over plain LoRA alone

Soft Prompt Tuning

Liu et al. [8] **What it is:**

- $m = 64$ learnable embedding vectors $\{p_1, p_2, \ldots, p_{64}\}$ prepended to the input token sequence
- Updated via gradient descent during training — **no manual prompt crafting required**
- At inference, the same learned tokens are prepended before every input

Why we use it:

- Manual prompts (e.g., "You are a gastroenterologist...") are brittle and cannot encode fine-grained domain nuances
- Learnable tokens capture **implicit domain knowledge** — endoscopic terminology, physician report style, grading conventions — purely through the training signal
- The tokens guide the LLM's attention toward the clinical domain *before* image and text tokens are processed
- Adds only $64 \times 4096 \approx 262$K parameters — negligible overhead with significant domain alignment benefit

Selective LoRA

Hu et al. [7] **What it is:**

- Low-rank adapter matrices injected into weight matrices: $W' = W + \Delta W = W + AB$
- $A \in \mathbb{R}^{d \times r}$, $B \in \mathbb{R}^{r \times k}$, rank $r = 64$, scaling $\alpha = 32$, dropout $= 0.05$
- Only A and B are trained; original W is frozen

Why SELECTIVE (not all layers):

- Transformer layers **specialise** at different levels of abstraction:
  - **Early layers** — low-level syntax and token patterns; minimal domain shift needed
  - **Middle layers** — semantic composition and concept associations; **critical for medical terminology**
  - **Last layers** — task-specific output formatting and clinical report style
- Targeting only these three groups **captures all domain adaptation needs** while training far fewer parameters
- Result: only **0.343%** trainable parameters vs. $\sim$1% for full LoRA — same quality, faster training, lower GPU memory

Why P-LoRA Outperforms LoRA Alone

Hu et al. [7], Liu et al. [8] **Plain LoRA — the limitation:**

- Adapts weight matrices but does not change how the model *interprets the input context*
- For highly specialised domains like GI endoscopy, the model still "reads" input through a generic lens — producing suboptimal attention over clinical features

P-LoRA — the combination advantage:

- **Prompt tokens** shift input representations toward the endoscopic domain *before* attention is computed
- **LoRA adapters** then operate on already-domain-aligned representations — more efficient weight updates with less training signal wasted
- The two mechanisms are **orthogonal**: prompt tuning acts at the input embedding level; LoRA acts at the weight matrix level

Empirical gains over plain LoRA:

- **+5% CIDEr** and **+4.4% ROUGE-L** on the GMC Calicut validation set
- GPU memory: ∼43 GB (comparable to plain LoRA ∼44 GB)
- Inference: ∼7 s/report on A100 80GB; deployable on RTX 3090 (24 GB)

System Architecture

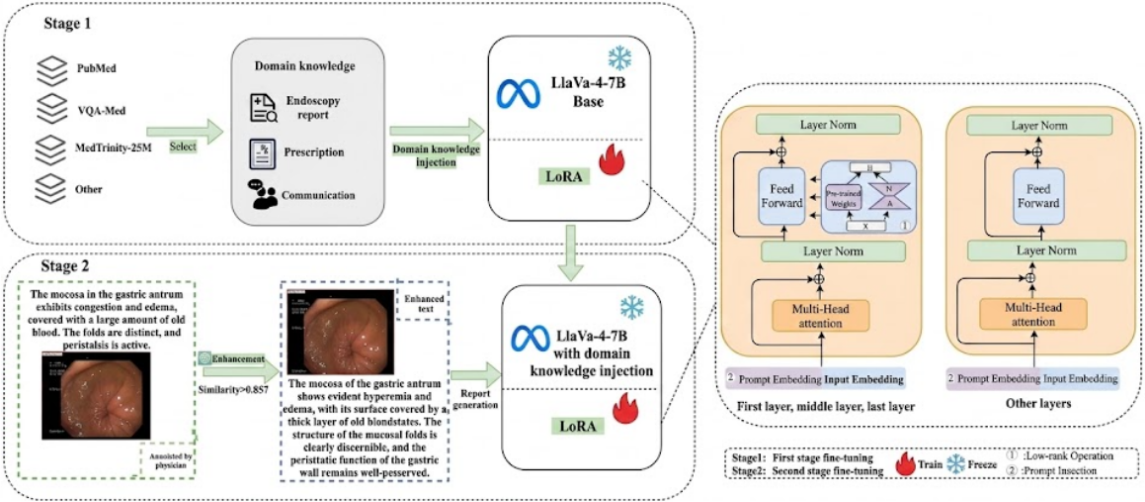


Figure: High-level architecture of the proposed vision-language model for endoscopic report generation.

Model Architecture – LLaVA

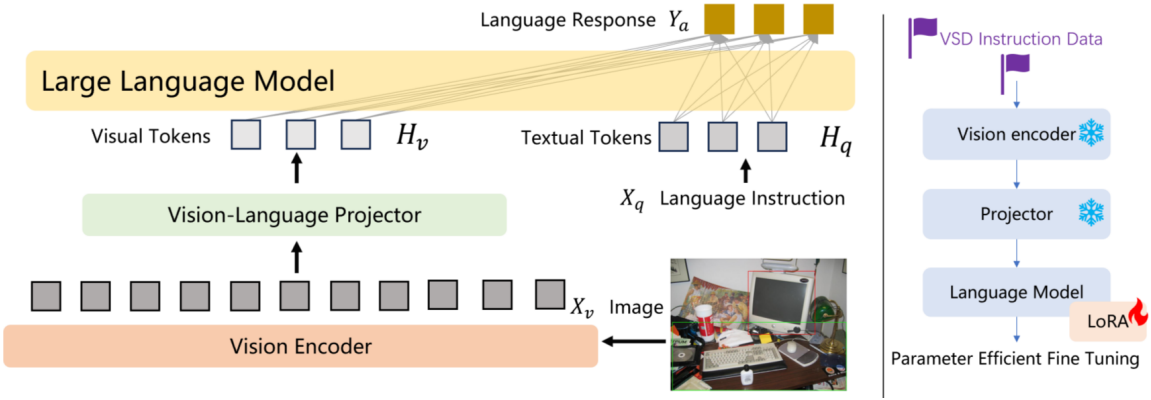


Figure: LLaVA architecture: vision encoder + projection layer + language model for multimodal understanding.

CLIP-ViT Architecture

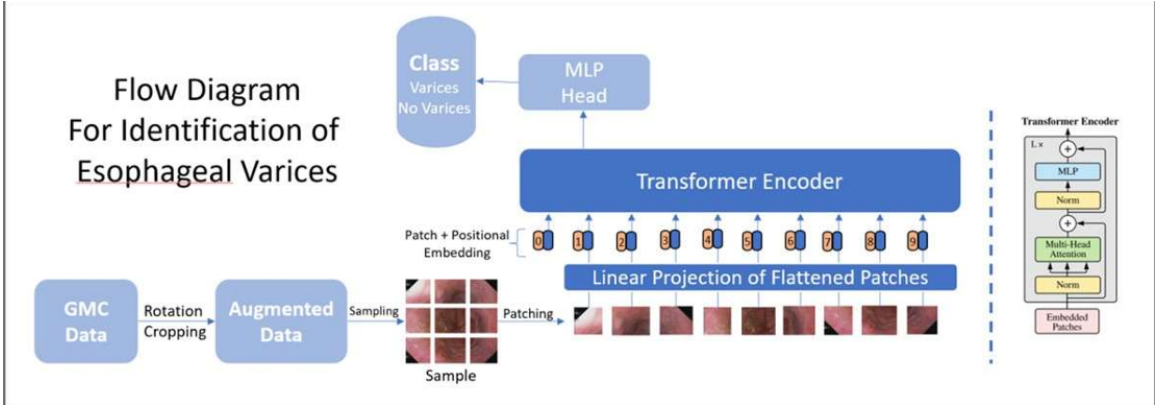
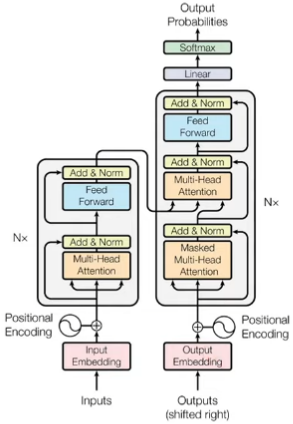
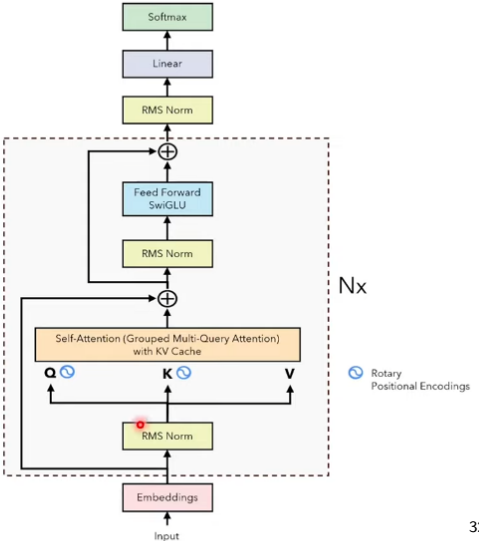


Figure: CLIP-ViT-L/14 vision encoder: patch embedding, transformer layers, and projected visual features.

Transformer vs LLaMA





Implementation

Refining Generation: Implementing System Prompts

Initial Challenge:

- Early model iterations produced verbose, unstructured, and redundant reports that did not match standard clinical formatting.

Implemented Solution (Dataset Reformatting):

- Integrated rigorous **System Prompts** directly into the training dataset.
- **Role Assignment:** Instructed the model to assume a specific clinical persona (e.g., "You are an expert gastroenterologist...").
- **Output Constraints:** Provided explicit instructions to prioritise direct, concise, and structured answers mirroring real GMC Kozhikode reports.

Result: A massive reduction in unnecessary text, yielding highly concise and structured outputs; further refinement improved diagnostic accuracy.

Qualitative Impact: Verbose vs. Concise Reports

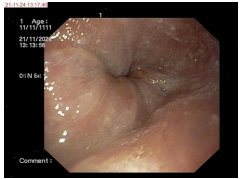
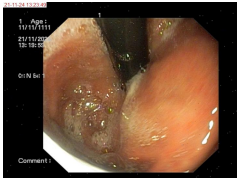
Previous Generation
(Verbose & Redundant)

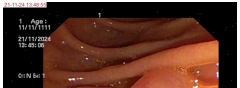
Medical Endoscopy Report

Patient Information

Patient Name: ABDUL MANAF
Patient ID: ABDUL MANAF_86376_RP241121_015818
Report Date: 2025-12-04 12:55:59

Endoscopy Images



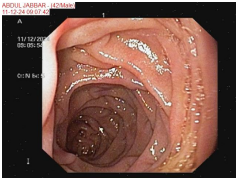
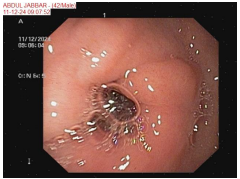
Current Generation
(Concise via System Prompt)

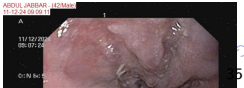
Medical Endoscopy Report

Patient Information

Patient Name: ABDUL JABBAR
Patient ID: ABDUL JABBAR_84105_RP241211_091205
Report Date: 2026-03-06 13:45:39

Endoscopy Images



Qualitative Impact: Verbose vs. Concise Reports

Previous Generation
(Verbose & Redundant)

Scope Passed Upto:	D2 mucosa without ulcers/ erosions/ luminal narrowing .Scope passed upto D3 mucosa without any difficulty .Biopsy taken from D2 and D3 mucosa for HPR and CBNAAT from D2 and D3 mucosa . D1 mucosa was edematous and erythematous with superficial ulcers . D1 was not entered further due to poor patient tolerability . D1 mucosa was edematous and erythemat
Premedication:	10% XYLOCAINE ANTRUM AND PYLORUS WITH SURROUNDING NODULARITY AND D1 NODULARITY FOR BIOPSY TAKEN RUT: PLAN: CECT THORAX + ABDOMEN AND BARIUM SWALLOW IF FURTHER EPISODE/HB FALL BIOPSY TAKEN FROM NODULARITY AND D1 NODULARITY FOR CBNAAT TAKEN RUT: PLAN: CECT THORAX + ABDOMEN AND BAR
Esophagus:	Normal mucosa without ulcers/ erosions/ luminal narrowing with normal peristalsis and distensibility noted .No foreign body seen in the esophagus .No esophageal varices seen .No altered blood seen in the lumen .No luminal blood seen in the esophagus .No foreign body seen in the esophagus .No luminal blood seen in the lumen .No luminal blood seen in the esophagus .No foreign body seen in the esophagus .No altered blood seen in the lumen .No luminal blood seen
Ge Junction:	40CMS and edematous and erythematous mucosa noted.Biopsy taken .No nodularity or ulcerations noted. No tear noted at present. No salivary pooling. No difficulty in swallowing. No foreign body seen. No luminal blood seen. No active ooze seen. No hemoptysis. No melena. No hematemesis. No hematuria. No proteinuria. No renal papilla. No ureter. No ampulla. No pyloric opening. No duodenal ulcer. No
Stomach:	No stasis . Normal peristalsis and distensibility noted . No ulcer/ erosions noted . Normal blood in lumen . No active bleed at present . Normal mucosal folds and contours noted . No luminal blood noted . Normal mucosal mucosa noted . No luminal blood noted . Normal mucosal mucosa noted . No luminal blood noted . Normal mucosal mucosa noted . No luminal blood noted . Normal mucosal
Fundus:	Normal mucosa without ulcers/ erosions/ luminal narrowing with normal distensibility and peristalsis noted .No food or fluid stasis noted .Normal distensibility and peristalsis noted .No food or fluid stasis noted .Normal distensibility and peristalsis noted .No food or fluid stasis noted .Normal distensibility and peristalsis noted .No food or fluid stasis noted .Normal

Current Generation
(Concise via System Prompt)

Stomach:	
Fundus:	No varices
Body:	Mosaic pattern of mucosa seen
Antrum:	Normal
Pylorus:	Normal
Duodenum:	D1: Normal | D2: Normal
Rut:	Mosaic pattern of mucosa seen
Biopsy:	Not taken
Impression:	GRADE II X3 COLUMNS OF ESOPHAGEAL VARICES WITHOUT RCS SEVERE PHG

Report generated by Medical LLaVA AI on 2026-03-06 at 13:45:39

The “Blackbox” Problem & Our Solution

The Initial Blackbox:

- The model would output a final diagnosis directly from an image without revealing its reasoning, making it hard to trust or debug.

Our Solution – Chain-of-Thought Integration:

- Augmented the training Q&A pairs with **step-by-step reasoning chains**.
- Each chain forces the model to structure its answer as: **anatomical identification** $\rightarrow$ **visual observation** $\rightarrow$ **diagnosis**.
- During training, the model learns to mimic this transparent reasoning process.

Outcome: The model now explicitly explains *what it sees* and *why it concludes something*, turning a black-box predictor into an **interpretable diagnostic assistant**.

Chain-of-Thought (CoT) in Endoscopy

Before CoT Implementation (Baseline Model):

- **Input:** [Image] "What is the diagnosis?"
- **Output:** "Grade II Esophageal Varices." *(correct but no justification)*

After CoT Implementation (Our Model):

- **Input:** [Image] "Analyze the image step-by-step and provide a diagnosis."
- **Output:**
  1. **Observation 1:** I observe the lower third of the oesophagus.
  2. **Observation 2:** There are swollen, tortuous submucosal veins present.
  3. **Observation 3:** The veins occupy less than 1/3 of the lumen, with no red colour signs.
  4. **Conclusion:** Therefore, the diagnosis is Grade II Esophageal Varices.

Impact: Explicit reasoning enhances clinical trust, allows verification, and helps identify subtle mistakes – a crucial feature for real-world deployment.

Fine-Tuning Configuration Details

Vision Encoder: CLIP-ViT-L/14

- Image resolution: 336×336 pixels
- Patch size: 14×14 (24×24 grid)
- Total vision tokens: 577 per image (576 patches + CLS)

Training Hyperparameters:

- **Precision:** bfloat16 (BF16)
- **Optimizer:** AdamW with 8-bit quantization
- **Learning Rate:** $2e^{-4}$ with cosine scheduler, warmup 0.1
- **Batch Size:** 2 micro-batch, 4 accumulation steps = 8 global
- **Attention:** Flash Attention 2
- **Epochs:** 4 epochs (both stages)

Training Configuration Rationale

- **CLIP-ViT-L/14** – pre-trained on 400M pairs; robust medical visual features; L size balances capacity vs. speed; 14×14 patch grid captures both anatomy and small lesions.
- **336×336 resolution** – standard for LLaVA; sufficient detail without excessive memory.
- **577 vision tokens** – each patch token carries localised information, enabling the language model to "see" global and local structures.
- **bfloat16** – halves memory vs. FP32, no performance loss.
- **AdamW + 8-bit** – stable weight-decoupled convergence; 8-bit reduces optimizer memory 75%.
- **LR $2e^{-4}$, cosine + warmup** – low peak prevents catastrophic forgetting; warmup stabilises early steps.
- **Batch size 8 (accumulated)** – simulates larger batch under GPU memory limits.
- **Flash Attention 2** – memory-efficient exact attention; 5-10× VRAM reduction for long sequences.
- **4 epochs** – sufficient for P-LoRA adapters on the augmented second-stage dataset without overfitting.

Why AdamW & Flash Attention 2?

AdamW over Adam

- Adam couples weight decay with gradient scaling – can over-regularize.
- AdamW **decouples** weight decay from adaptive gradients → better generalization.
- Standard for LLaMA fine-tuning; prevents over-shrinking of LoRA adapters.

Flash Attention 2 over regular / Flash Attention 1

- Standard attention: $O(N^2)$ memory – prohibitive for 577 vision tokens + text.
- Flash Attention 1 tiled computation, but under-utilised GPU parallelism.
- Flash Attention 2:
  - **Fewer non-matmul operations** → higher throughput.
  - **Better batch/head parallelism** → faster training.
  - **Lower VRAM usage (5-10× reduction)** → feasible on single GPU.

Together: Stable convergence + memory-efficient training at scale.

Results

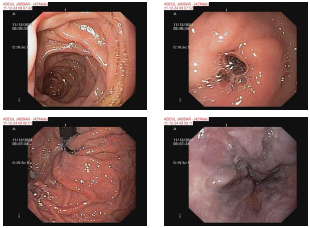
Generated Report

Medical Endoscopy Report

Patient Information

Patient Name:	ABDUL JABBAR
Patient ID:	ABDUL JABBAR_84105_RP241211_091205
Report Date:	2026-04-13 18:09:26

Endoscopy Images



Findings

Indication:	CLD-UGIB
Scope Passed Upto:	D2
Premedication:	Lignocaine spray
Esophagus:	Normal
Ge Junction:	Lax at 38cm
Stomach:	
Fundus:	Edematous mucosa noted
Body:	Edematous mucosa noted
Antrum:	Erythematous mucosa noted
Pylorus:	Normal
Duodenum:	D1: Normal | D2: Normal
Rut:	Not taken
Biopsy:	Not taken
Impression:	GRADE II ESOPHAGEAL VARICES WITH RCS EVL DONE SEVERE PHG

Report generated by Medical LLaVA AI (FFT) on 2026-04-13 at 18:09:26

Final diagnostic report – Pages 1 and 2

Evaluation Metrics

CIDEr
Consensus-based evaluation using TF-IDF weighted n-gram vectors.

$$\text{CIDEr}(c, S) = \frac{1}{M} \sum_m \frac{\mathbf{g}^n(c) \cdot \mathbf{g}^n(s_m)}{\|\mathbf{g}^n(c)\| \, \|\mathbf{g}^n(s_m)\|}$$

ROUGE-L
F-measure based on longest common subsequence (LCS).

$$\text{ROUGE-L} = \frac{(1+\beta^2)\, R_{lcs} \cdot P_{lcs}}{R_{lcs} + \beta^2 P_{lcs}}$$

METEOR
Combines precision, recall, and a fragmentation penalty.

$$F_{mean} = \frac{10PR}{R + 9P}, \quad \text{METEOR} = F_{mean}(1 - \text{Pen})$$

BLEU-4
Geometric mean of 1–4 gram precisions with brevity penalty.

$$\text{BLEU-4} = \text{BP} \cdot \exp\left(\frac{1}{4} \sum_{n=1}^{4} \log p_n\right)$$

BERTScore
Semantic similarity via contextual BERT token embeddings.

$$F_{\text{BERT}} = \frac{2\, P_{\text{BERT}} \cdot R_{\text{BERT}}}{P_{\text{BERT}} + R_{\text{BERT}}}$$

All five metrics reward similarity between generated and reference reports; higher is better (↑).

Quantitative Comparison — MVLMERG vs. Baselines

Method	CIDEr↑	ROUGE-L↑	METEOR↑	BLEU-4↑	BERTScore↑
Qwen-VL-7B	0.0212	0.2815	0.2369	0.2293	0.6947
VisualGLM-6B	0.0147	0.2210	0.1913	0.1829	0.5893
GPT-4V	0.0955	0.3040	0.3311	0.3292	0.7846
HuatuoGPT-Vision-7B	0.0802	0.2642	0.2791	0.2709	0.7538
LLaVA-Med	0.0171	0.1979	0.2677	0.2417	0.7452
BiomedGPT	0.0233	0.1739	0.2381	0.2178	0.7117
Uni-Med	0.0881	0.2013	0.1928	0.1903	0.6639
EndoChat	0.0578	0.1965	0.2102	0.1961	0.6740
MVLMERG (Ours)	**0.2731**	**0.4621**	**0.4052**	**0.3902**	**0.8244**

MVLMERG achieves state-of-the-art on all 5 metrics ($p < 0.01$, Cohen's $d > 0.8$). Report accuracy: **85.19%**.

Expert Physician Evaluation

Ali et al. [4] **Evaluation Protocol:**

- 2 senior endoscopists (10+ years experience), GMC Calicut
- Each generated report labelled **Correct / Incorrect** based on clinical accuracy
- Disagreements resolved by **joint review and consensus discussion**

Method	**Accuracy** $\uparrow$
HuatuoGPT-Vision-7B	0.7240
LLaVA-Med	0.7013
BiomedGPT	0.6426
Qwen-VL-7B	0.5659
EndoChat	0.6734
GPT-4V	0.7833
MVLMERG (Ours)	**0.8519**

Calibration Analysis

Why calibration matters in clinical AI:

- A model can be accurate yet **overconfident** — in clinical settings, miscalibrated confidence can mislead physicians who rely on model certainty to decide when to trust or override the output
- **ECE (Expected Calibration Error)** measures the gap between predicted confidence and actual accuracy across confidence bins

MVLMERG Calibration Results:

- ECE = **0.042** — indicating **well-calibrated** predictions
- Token-level confidence binned into 10 equal intervals
- Confidence aligns closely with actual correctness: when the model says 80% confident, it is correct ∼80% of the time
- **Not overconfident** — clinicians can reliably use the confidence signal

Inference Speed:

- ∼**7 s/report** on A100 80GB vs. 3–5 min manual writing — **26**× **faster**
- Deployable on RTX 3090 (24 GB) via quantization

Hardware & Training Setup

Training Hardware:

- **GPU:** NVIDIA A100 80GB PCIe (CUDA 12.9)
- **CPU:** Intel Xeon Silver 4310 @ 2.10 GHz (24 physical cores, 48 threads, 2 sockets)
- **RAM:** 125 GiB
- **Python:** 3.10.12; PyTorch (BF16 + Flash Attention 2)

Training Resource Usage:

- GPU memory during training: $\sim$43 GB
- Stage 1 training time: $\sim$70 h
- Stage 2 training time: $\sim$15 h
- Trainable parameters: **0.343%** of full model

P-LoRA Hyperparameters:

- LoRA rank $r = 64$, $\alpha = 32$, dropout $= 0.05$
- Prompt tokens $= 64$
- Epochs: 4 (both stages)
- Seed: 42
- Max sequence length: 4096

Minimum Deployment Spec:

- 16 GB RAM, Python 3.8+
- CUDA 11.2+, PyTorch 2.0+
- Can run on RTX 3090 (24 GB)
- $\sim$7 s inference latency

Future Work

Model Optimization & Quantization

GGUF Format Conversion Dettmers et al. [15], Frantar et al. [16]

- Optimizes CPU and mixed CPU-GPU inference
- Significantly improved model loading times Frantar et al. [16]

Multi-Level Quantization Strategy Dettmers et al. [15], Frantar et al. [16]

Format	Use Case	Size Reduction
Q8_0	High-end workstations	∼50%
Q5_K_M	Standard clinical computers	∼69%
Q4_K_M	Basic clinical terminals	∼75% (3.25GB) Dettmers et al. [15]
Q3_K_L	Mobile deployments	Ultra-compact

Deployment Planning

Two Deployment Strategies:

1. **vLLM Framework** Kwon et al. [17]
   - High-throughput clinical environments
   - Concurrent access scenarios
2. **Ollama Framework** Kwon et al. [17]
   - Individual workstations
   - Low-resource environments
   - Resource-constrained deployment

EHR Integration:

- RESTful API architecture (secure, authenticated endpoints)
- Real-time synchronization with patient records

Conclusion

Key Achievements:

- Built a vision-language model for automated endoscopic report generation, fine-tuned on a custom ∼31,200-sample dataset.
- Implemented a DPDP-compliant de-identification pipeline, making sensitive clinical data safe for AI training.
- Integrated system prompts and Chain-of-Thought reasoning to produce **concise, transparent, and clinically interpretable** reports.
- Achieved state-of-the-art: CIDEr 0.2731, ROUGE-L 0.4621, METEOR 0.4052, BLEU-4 0.3902, BERTScore 0.8244, report accuracy **85.19%**.

Impact: Drastically reduces documentation time; provides explainable AI assistance to gastroenterologists Shanafelt et al. [1], Rotenstein et al. [2]. **Path Forward:** Model quantization Dettmers et al. [15], Frantar et al. [16] and clinical deployment trials (pending IEC approval) will bring the tool into real-world endoscopic workflows.

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